Pakhi Banchalia

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Education

Technische Universität Dresden 2025 – Present
MSc in Computational Modeling and Simulation, Applied AI track (3rd Semester)

Amrita Vishwa Vidyapeetham, Amritapuri 2021 – 2025
Bachelor of Technology in Computer Engineering, Honors in Artificial Intelligence

Experience

Summer Research Intern, Aalto University Jun 2026 – Aug 2026
Espoo, Finland
- Selected as one of **67 interns from 5,500 applications** across 133 countries for the AScI International Summer Research Programme, working on **empowerment-based assistance in multi-agent reinforcement learning**.
- Found that AvE (NeurIPS 2020) tests assistance with an agent acting remotely rather than embodied; rebuilt the experiment with an embodied assistant to measure what that costs.

Research Intern (Independent Collaboration), CMC Lab, TU Dresden Nov 2024 – Sep 2026
Dresden, Germany (Intermittent)
- Designed RL agents with **internal physiological drives** to study retention and generalization across changing task distributions.
- Analysed neural and behavioural data from trained foraging agents, decoding movement and internal state from network activity and linking oscillating unit groups to gait.

Research Intern, Lossfunk Jul 2025 – Sep 2025
Bangalore, India
- Built and evaluated **homeostatic RL agents** in **Crafter** driven by hunger, energy and safety, and explored intrinsic drives such as **curiosity** to support exploration alongside survival.
- Tested how reward formulations and competing drives affect exploration, survival, behavioural diversity and policy stability, against standard RL baselines.

Contributor, Google Summer of Code (INCF) GitHub May 2024 – Sep 2024
Remote
- Built a **growing graph Neural Cellular Automaton** of the *C. elegans* embryo that adds each cell as it divides and learns where daughter cells sit in 3D, trained with **CMA-ES evolutionary optimization**.
- Mapped cell-by-cell embryo data with **Topological Data Analysis** (Kepler Mapper) into interactive visualizations, released as documented open-source code.

Core Committee Member; Design Lead, amFOSS Sep 2022 – Jul 2025
Amritapuri, India
- Led **amFOSS**, a **student-run open-source software community**, as a core committee member: ran weekly project check-ins, caught stalled work, and was the link between the founder, university staff and members.
- Headed the **design team**, reviewing work and making final design decisions, including the **amfoss.in** redesign.

Projects

Polaris: Rare-Disease Atlas Live GitHub 2026
- Built a rare-disease knowledge graph at the **Hack-Nation Global AI Hackathon** (Buffalo Initiative and OpenAI challenge), mapping patient communities by mechanism and mutation direction, every link sourced.
- Co-developed the idea, then built the interface and researcher view, which screens 2,868 approved medicines by mutation direction.

Alignd: AI Sponsorship Agent GitHub 2026
- Selected as **one of 60 demos from 2,000+ applications** to the TinyFish Accelerator, a $2M program run by TinyFish and Mango Capital, and reached **Phase 2**.
- Built an AI agent that finds and vets YouTube creators for a brand's campaign in six stages, deciding what the agent may conclude alone and what the brand can overrule.

MedX: X-ray Editing Tool 2024

- Reached the **final round of GE HealthCare's Precision Care Challenge 2024** with a web tool that edits X-ray images to generate training data.
- Owned the Next.js and TypeScript front end: visual design, layout, and the canvas where a user places an object into an X-ray.

Hospital Inventory System 2022 – 2023

- Deployed a Django inventory-management system for Amrita Hospital as part of a team, built to the hospital's requirements.
- Focused on building the ticketing and activity workflow for creating, assigning, tracking and resolving issues, plus UI work and bug fixes.

Publication

CAN: Continuously Adapting Network [Springer, ICDSA 2025] Link 2025

- Proposed a **continual-learning network** that learns new tasks without forgetting old ones by freezing task-critical neurons and applying **Hebbian learning**.
- Extended CAN in my B.Tech. thesis, **SEANN (Self-Expanding and Adaptive Neural Networks)**, so the network **grows its own capacity** and finds task-relevant subnetworks.

Talks and Posters

Emergence of Brain-like Features in Homeostatic RL Agents Link 2025

- Accepted as an **oral talk at BrainBody 2025**; co-authored work showing how internal states, neural dynamics and adaptive exploration are coupled.

Brain-like Features in Homeostatic Deep RL Agents Link 2025

- Accepted as a **poster at Bernstein Conference 2025**; co-authored work showing that agents combining internal and sensory inputs develop structured behavioural and neural patterns.

Skills

Programming Languages: Python, C++, C, Java, JavaScript, TypeScript, SQL, MATLAB, HTML, CSS
Machine Learning and AI: PyTorch, TorchVision, Gymnasium, JAX, Reinforcement Learning, Multi-Agent Reinforcement Learning, Continual Learning, Graph Neural Networks, Neural Cellular Automata, AI Agents, LLM APIs, Anthropic API, Gemini API
Data Science: NumPy, SciPy, Pandas, Matplotlib, Kepler Mapper, Weights and Biases
Software Development and Design: FastAPI, Django, Next.js, React, n8n, Git, Linux, Docker, Figma
Languages: English (C1); German (A2)